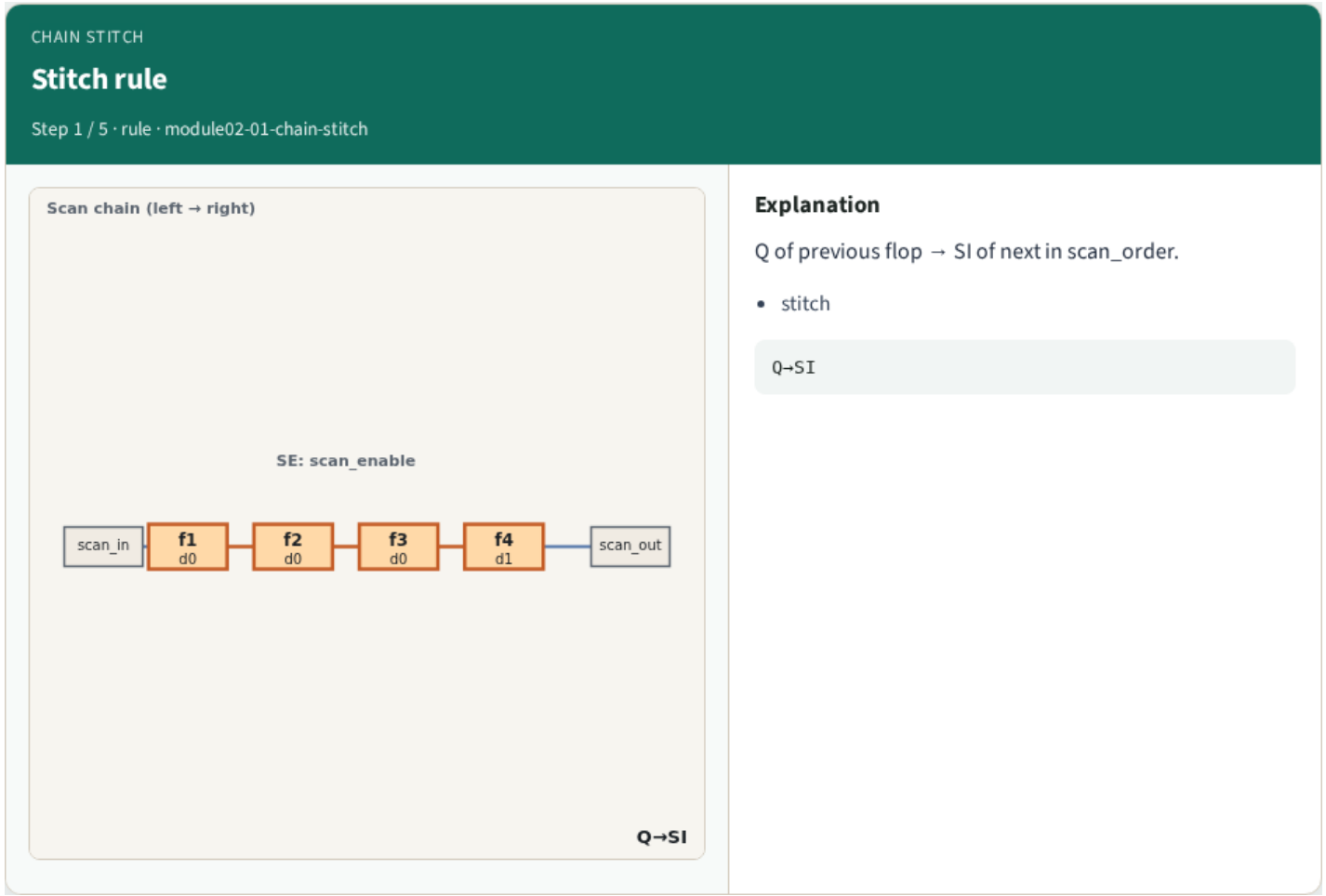
Why stitch

Chain stitch wires Q of previous flop to SI of next in scan_order

The idea

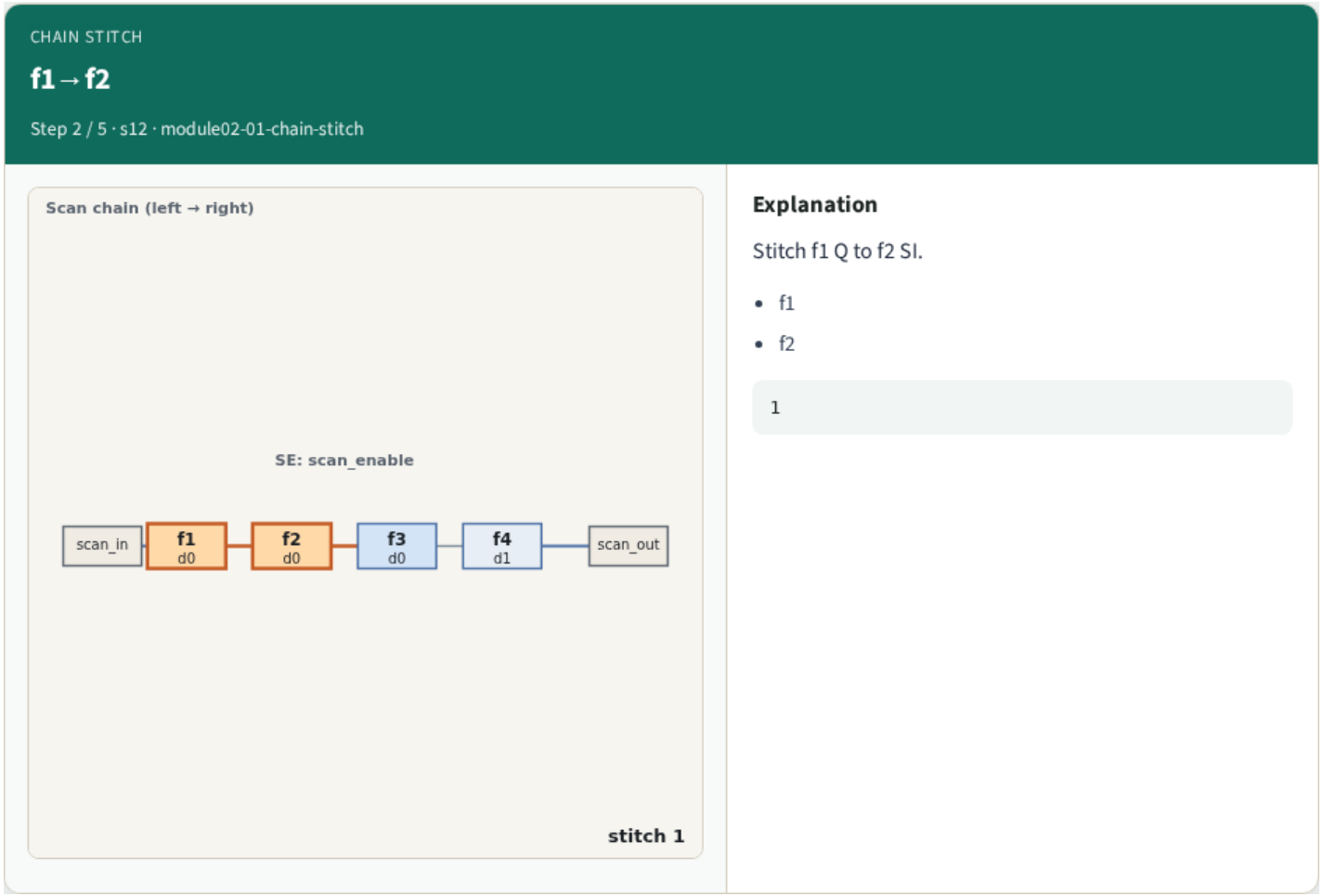
- GOLDEN: stitch_count=3

Stitch rule



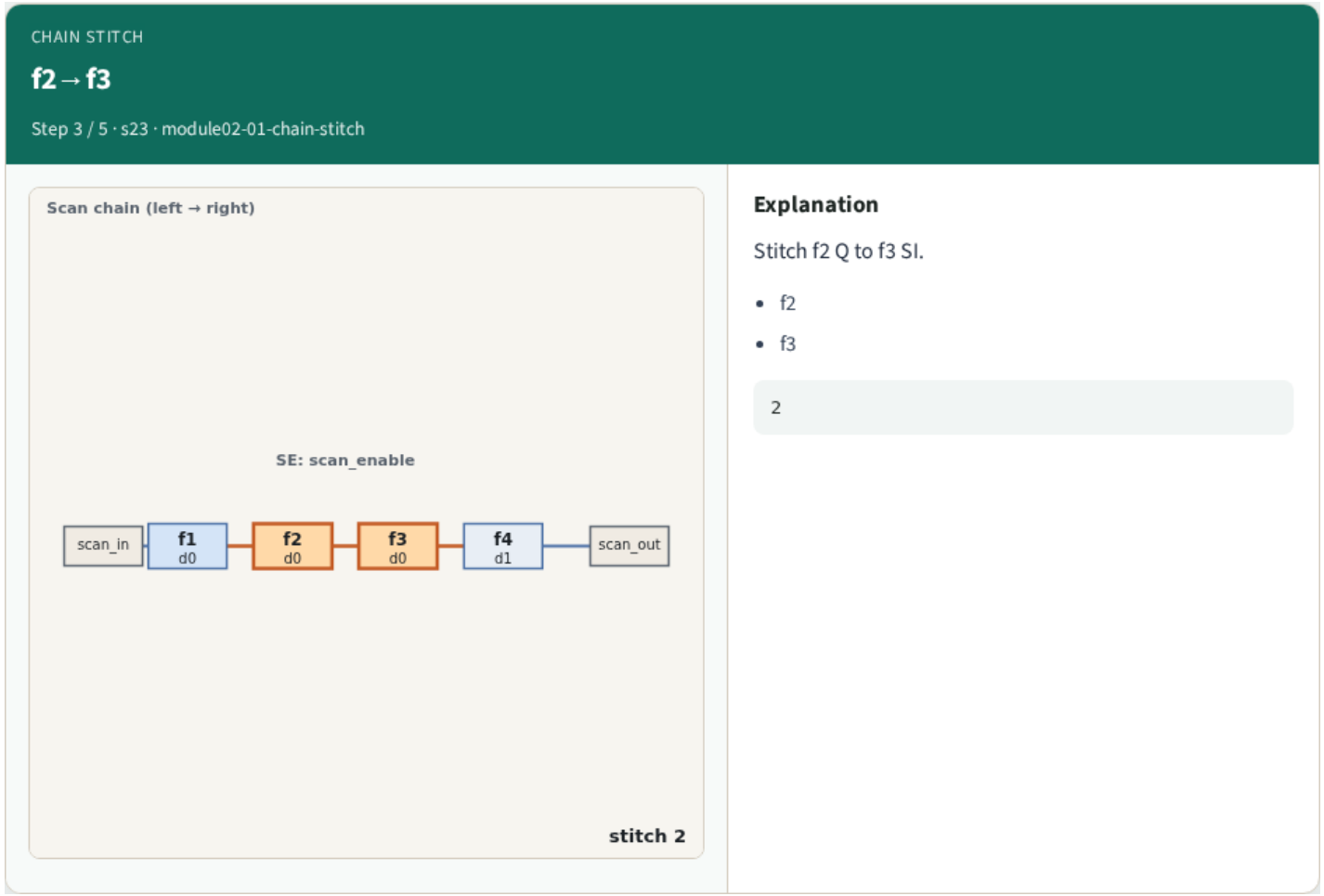
Stitch rule

f1→f2


CHAIN STITCH
f1 → f2
Step 2 / 5 · s12 · module02-01-chain-stitch
Scan chain (left → right)
SE: scan_enable
scan_in
f1
d0
f2
d0
f3
d0
f4
d1
scan_out
stitch 1
Explanation
Stitch f1 Q to f2 SI.
f1
f2
1

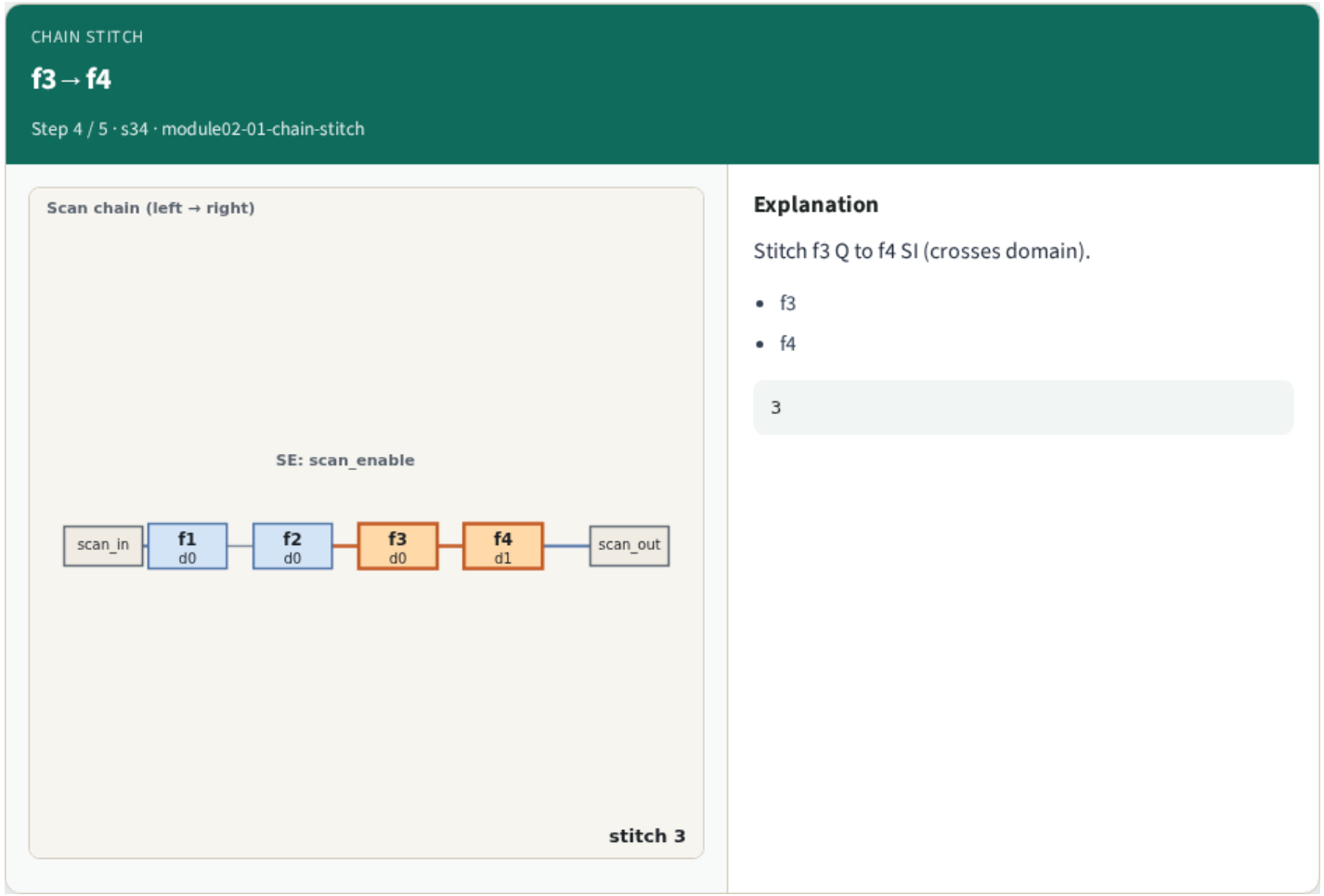
f1→f2

f2→f3


CHAIN STITCH
f2 → f3
Step 3 / 5 · s23 · module02-01-chain-stitch
Scan chain (left → right)
SE: scan_enable
scan_in
f1
d0
f2
d0
f3
d0
f4
d1
scan_out
stitch 2
Explanation
Stitch f2 Q to f3 SI.
f2
f3
2

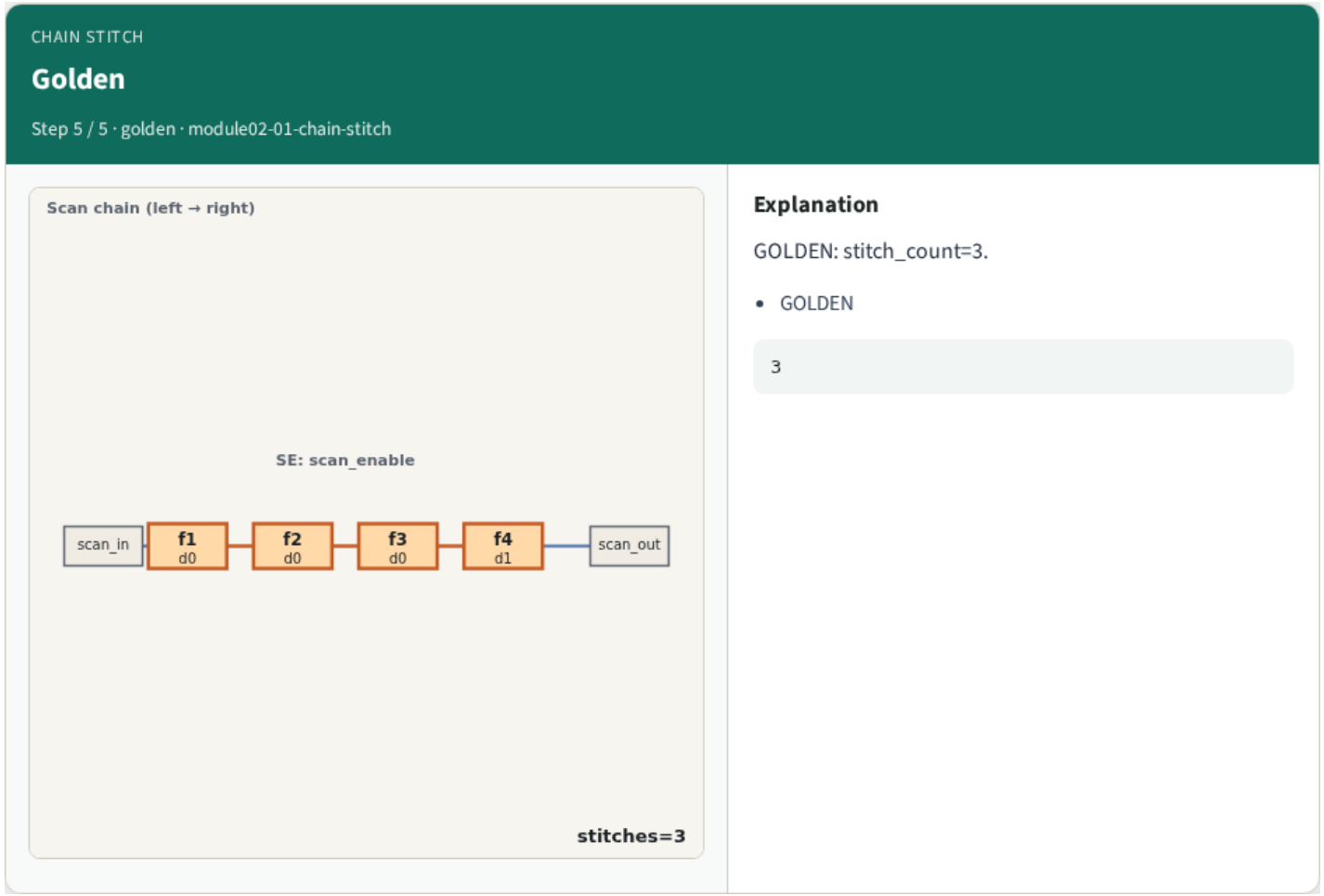
f2→f3

f3→f4


CHAIN STITCH
f3 → f4
Step 4 / 5 · s34 · module02-01-chain-stitch
Scan chain (left → right)
SE: scan_enable
scan_in
f1
d0
f2
d0
f3
d0
f4
d1
scan_out
stitch 3
Explanation
Stitch f3 Q to f4 SI (crosses domain).
f3
f4
3

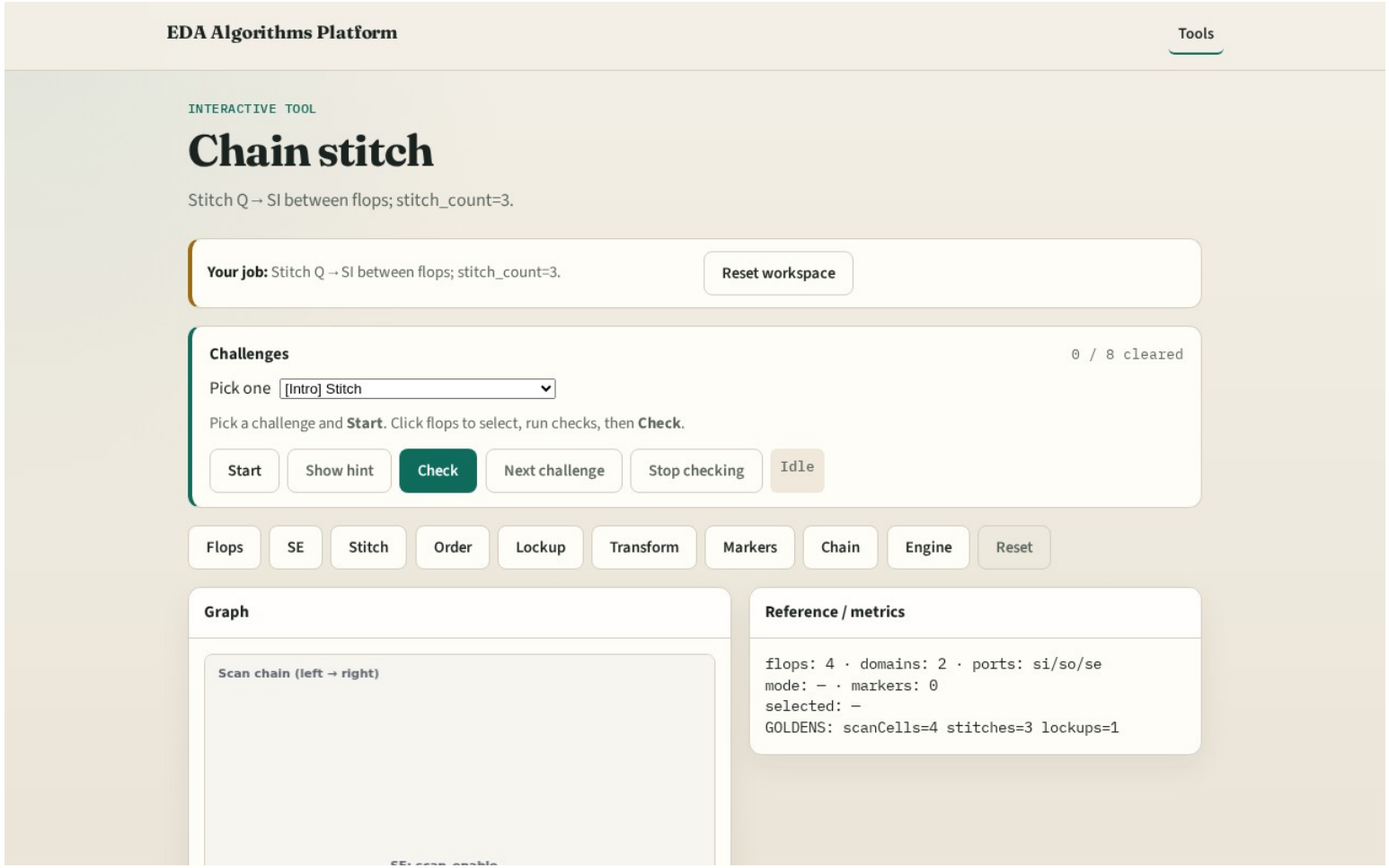
f3→f4

Golden



Golden

Browser lab track



Browser lab starter

Implement track

- Call stitch_chain(); assert stitch_count==3

Pitfalls

- Counting SI/SO hooks as stitches
- Wrong order

Your turn

- Next: scan order check